



Lane-level routable digital map reconstruction for motorway networks using low-precision GPS data

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ABSTRACT

The construction of routable digital maps based on trajectory data has attracted a lot of attention, especially in recent years, with the ease and cheapness of collecting the required data. Such maps, if they are constructed at lane-level, have many applications in traffic analysis, especially the study of driving behavior based on floating car data. In this paper, we present a three-step automatic method based on QuickBundles for the node detection in the road network, a dissimilarity matrix based on Fréchet distance for road centerline construction, and the Gaussian Mixture Method for lane estimation. The results are a smooth, segment-based centerline unbiased by GPS density distribution over lanes with accurate road width as well as compatible and highly accurate estimation of lanes. The accuracy, connectivity, compatibility, validity, and robustness of the proposed method have been tested in various ways. The results of this paper show that this method, while being low cost, can construct accurate lane-level routable digital maps that can be used as a platform for extracting longitudinal and lateral driving behavior, especially drivers' lane-changing maneuvers. Due to the unique features of the proposed method, the width of the lanes remains constant along the entire length of the road segments. In addition, the validation of the method based on traffic metrics shows that the constructed maps can be used to obtain reliable estimates of the speed and volume of traffic flow in different lanes. Being derived from actual trajectories, the inferred lane markings may deviate from the physical ones if the driver population systematically deviates laterally, for instance in curves; depending on the application use case of the resulting map, this may be a desirable or undesirable feature.

1. Introduction

It has always been important to prepare a detailed road map for various practical and academic purposes. With the development of Intelligent Transportation System (ITS) applications a new concept has emerged as Routable Digital Maps (RDM). These maps are connected and directed graphs, with features such as travel time, speed limit, etc. which allow routing and other usages. Nowadays, many users use RDMs to find the shortest path between their origin and destination and to avoid traffic congestion. In recent years, numerous efforts have been made commercially (such as Here by Nokia and Garmin) that usually require a paid subscription and are usually not available for research purposes, and also non-commercially (such as Google Maps, OpenStreetMap, and Waze) to develop RDMs. Some academic research has also focused on the concept of map construction, which aims to make RDMs as accurate as possible. GPS data, which has become much cheaper to collect thanks to smartphones, have greatly accelerated the development of RDMs. Using

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GPS data to develop RDMs is known mostly as Automated Map Construction (AMC). One important issue that is not well addressed in the literature so far is RDMs that have lane information.

The main motivation of this paper is preparing a dataset for lane-based traffic analysis based on trajectory data. Although there are many examples of data and studies of longitudinal behavior of drivers (car-following studies), very few datasets exist on lane change behavior, and developing lane-based RDMs is a first step to enabling us to extract lane change behavior from trajectories. Such a map helps us to significantly reduce the lateral bias of the trajectory data. Also, the analysis of the traffic flow and the longitudinal and lateral maneuvers of the drivers will be done more accurately if it is based on a map that is fully compatible with the input data (trajectories). As a result, this map can be used as a platform for accurately extracting vehicles' lateral maneuver data (such as lane changing) and would be the first step toward collective traffic management strategies for notorious bottlenecks in motorway networks like merging or weaving sections. Besides, such a map may be used for autonomous vehicle navigation and online drivers' guidance. These last two are outside of the scope of the authors' motives. But as the potentials of the future use of the results of this paper should not be neglected.

The commercial importance of the development of lane based RDMs has led to a relatively limited number of scholarly papers in the literature. These efforts suffer from three major drawbacks. First, lanes are calculated on the basis of a centerline as the road skeleton. In most studies, this skeleton is derived from the density of trajectory points or paths. As a result, the change in the concentration of traffic flow from one side to the other side of a segment, which is relatively common at weaving sections, can bias the road skeleton at least in such segments. The second common drawback is that one-dimensional classification methods are commonly used for lane identification purpose. This can cause the width of a lane to not be constant throughout a segment of the road, while road designers usually maintain the width of the lanes constant through the segment. And finally, the third drawback is that these efforts are mainly done in the field of geographical studies. So almost always the benchmark used to validate the results was a reliable map as ground truth, whereas traffic benchmarks such as conformity of the proportion of vehicles in each lane based on identified lanes and loop detectors' data were never considered.

The main aim of this paper is to propose a low cost and high accuracy algorithm that is capable of reconstructing a lane based RDM for motorway networks based on the low-accuracy GPS trajectories collected by smartphones. What our paper shows is that by combining large amounts of low-precision data the underlying lane structure of the segments can still be discovered with reasonably high accuracy, despite undeniably large inaccuracy of every single data point. The data used in this study was collected on a test site around the Antwerp ring in Belgium. Our methodology consists of three main steps. Firstly, by identifying intersections within the test site, the motorway network is divided into segments. In the second step, the centerline of each segment is identified. The determination of this centerline is based on a dissimilarity matrix that is computed based on the Fréchet distance between trajectories in each segment. Consequently, the road centerline is smoothed and unbiased compared to density-based methods. Finally, in the third step, a two-dimensional clustering based on the Gaussian Mixture Method (GMM) is used to identify the road lanes. This causes the width of the lanes to remain constant throughout the road segment. In addition, the way we applied GMM ensures that the error caused by fixed obstacles (such as bridges, etc.) in the trajectory data does not cause bias in the estimated lanes. Being derived from actual trajectories, however, the inferred lane markings delineate the space that drivers use in practice while driving "in a lane". As a result, these functional lane markings may deviate from the physical ones if the driver population systematically deviates laterally, for instance in curves. Depending on the application use case of the resulting map, this may be a desirable or undesirable feature. We validate the results of our proposed method by comparing the speed of the identified lanes and the share of each lane from the volume of traffic flow in the identified lanes compare to the loop detectors' data. We used the normalized root-mean-square error (NRMSE) for validation. Two main limitations of this study are type of the road network to which this method is applicable and the matter of connectivity. This method is only performed on motorway segments and using and evaluation of the proposed method on urban roads is still an open area of study. The matter of connectivity in the proposed method is a compromise. In this study, the accuracy of lane estimation is more important to us. Consequently, the connectivity has been sacrificed to some extent in return for high accuracy lane estimation. This issue will be discussed in detail later in [Section 5.2](#).

The rest of this paper is organized as follows: section two is devoted to the literature review. We introduce the test site and collected data in section three. Our methodology is explained in section four and the lane identification results, as well as the validation and robustness analysis, are presented in section five. Finally, we conclude our study and propose some directions for further studies in the last section of the paper.

2. Literature review

Trajectory data can be collected from various sources such as radio frequencies, video recording, aerial imaging, GPS, laser sensors etc. Trajectory data has been used to reconstruct outdoor maps (road maps) and indoor maps ([Zhou et al., 2018](#), [Li et al., 2019](#)) (which is completely out of the scope of this article). In addition, a group of studies in this field have only identified nodes and intersections in the road network ([Deng et al., 2018](#), [Tang et al., 2019a](#)). It cannot be ignored that accurate identification of nodes is a very important step in reconstructing accurate RDMs. Among them, some have even identified traffic signals and phase duration data ([Ruhhammer et al. 2016](#)) or intersections' lanes ([Yang et al., 2018a](#)).

Methods for road map construction based on GPS tracks in the literature are usually classified based on the core algorithm that is used to construct the road centerline (the second step in this paper). Biagioni and Eriksson classify AMC algorithms into three groups and perform a qualitative, quantitative, and comparative evaluation. The three groups are: K-mean, kernel density estimation (KDE) and trace merge algorithms. They conclude KDE-based techniques, which are built on producing a kernel density estimation of the raw GPS points and are capable of dealing with massive amounts of GPS traces, are outperforming the others algorithms ([Biagioni and](#)

Eriksson 2012). A good example of KDE based algorithms that developed a score function to improve the accuracy of topology of identified nodes is proposed by (Kuntzsch et al. 2016). A very recent example of density-based algorithms with good performance demonstrated over several datasets is developed by (Fu et al. 2020). Ahmed et al. propose a different classification for AMC algorithms and perform some evaluation of different algorithms with similar datasets. They conclude AMC algorithms could produce either accurate maps, or maps with good coverage, but not both. They classified all point classification-based algorithms into one point-clustering category, covering both K-mean and KDE algorithms. They also introduced a third category named intersection linking, which is based on the identification of the intersection through a series of criteria (Ahmed et al. 2015a, Ahmed et al. 2015b). An example of using these criteria to identify the network's nodes is nodes identification based on the changes in the direction of movement of the trajectories (Karagiorgou and Pfoser 2012, Xie et al. 2015). Another study based on the intersection linking method uses a prototype detector trained on ground truth data from existing maps (Fathi and Krumm 2010). Alternative methods detect the longest common subsequences between each pair of GPS tracks and then partition the longest nonconsecutive subsequences into consecutive sub-tracks to reuse them as connecting points (Xie et al. 2017). Other methods use bindery circles to categorize an area into road segments and node districts (Wang et al. 2015). A DBSCAN-based point clustering method is also used to detect nodes in the network (Ezzat et al. 2018). In a study based on the trace merge algorithms, authors used travel time information as a criterion to identify which trajectory traces should merge (Bruntrup et al. 2005). In another study, the authors collected 20 million trajectory sample points from campus shuttles. They added a smoothing step before applying the main algorithm to improve the final accuracy of the results (Cao and Krumm 2009). The effect of a filter based on speed, acceleration, and maximum allowable turning angle has been analyzed to improve the accuracy of estimating the road network based on a trace merging algorithm (Niehoefer et al. 2009). And a modified version of Fréchet distance was proposed for more accurate partial matching of the trajectory traces (Ahmed and Wenk 2012). The reconstruction of digital maps using the GPS traces has also been used for the pedestrian networks. The method presented in this paper can be considered in the category of trace merging algorithms (Kasemsuppakorn and Karimi 2013). A combination of intersection linking, and trace merging is also proposed. In this method network nodes are identified based on the direction change of the GPS traces and then trajectories of each road segment precisely adjusted and merged through a greedy method (Xie et al. 2016). In addition to the above-mentioned classified methods, we can also mention machine learning techniques (Agamennoni et al. 2010) and structure learning techniques (Huang et al. 2018) based on the GPS data of the vehicles and Genetic algorithm based on GPS data collected by smartphones (Costa and Baldo 2015) to reconstruct road networks. Moreover, other methods are proposed that can detect changes in the road network based on a point-to-segment matching algorithm (Tang et al., 2019b) or partial curve matching method using Fréchet distance (Zheng and Zhu 2019). And then create new road segments and update the existing maps. Also, in order to investigate road horizontal alignment, a study used data collected by a probe vehicle equipped with GPS and an HD camera to reconstruct the road centerline. The main purpose of this research was to extract the road centerline including horizontal curves with high accuracy (Marinelli et al., 2017a).

One of the earliest attempts for lane identification based on low-precision GPS data was done by Wagstaff et al. They used a K-means algorithm for one-dimensional clustering of the lateral distances of the GPS tracks relative to a pre-defined centerline (Wagstaff et al. 2001). Among all the papers which are reviewed by (Biagioni and Eriksson 2012) and (Ahmed et al. 2015a) only one of them was able to identify road lanes. This paper used Bentley and Ottmann algorithm to transform GPS information into a graph structure and then applies a K-means clustering to identify the lanes (Edelkamp and Schrödl 2003). They further develop their algorithm by using a spatial clustering algorithm that guarantees the connectivity of lanes through the road segments (Schroedl et al. 2004). Because they project trajectory points on lines perpendicular to the centerline of the road segment, the resulting width of a reconstructed lane may not be constant throughout a segment of the road. The idea of perpendicular lines for one-dimensional classification of the trajectories was also used in (Chen and Krumm 2010) to identify the traffic lanes at three intersections in Seattle. For classification into lanes, they used the GMM method. Although authors claim to identify the lanes accurately, the use of perpendicular lines also has the risk of discontinuity of lane width along the road segment. A recent alternative for the previous classification methods for lane identification is a Naive Bayes classifier. The authors claim that this method can be used to update existing maps and add lane information to them based on low-precision GPS trajectories. However the auteurs still used perpendicular cross-sections of the trajectory points across the road, and so they reduced the classification dimension to one (Tang et al. 2015). In follow-up work, the same authors also examined the use of the one-dimensional GMM method in detecting lanes and compared the results with satellite images (Tang et al. 2016). The Kernel density estimation has been used to determine the centerline of the road lanes for three road types. The input data was low accuracy GPS data collected using a floating car (Uduwaragoda et al. 2013, Neuhold et al. 2017). Also, a fuzzy sets-based algorithm for limited applications of lane-level vehicles' positioning near signalized intersections using smartphones' GPS data has been proposed (Marinelli et al., 2017b). Yang et al. in their two-step fuzzy logic-based approach, first match the trajectory data with existing maps and then updated these maps based on the detected lanes (Yang et al., 2018b).

Alongside with working on low-precision GPS data, there are some other studies that used high-accuracy sources such as laser and Yaw rate sensors, cameras and high-precision GPS to reconstruct RDMs and identify lanes more accurate. One of the first studies in this field used the traffic surveillance cameras to collect trajectory data. In this study, vehicle tracking was performed using the Kalman filter and lanes identification was performed by K-means clustering method (Melo et al. 2006). Jimenez et al. equipped a probe vehicle with five sensors, including GPS, camera, and gyroscope. They reconstruct digital maps including highly accurate geometric curves based on collected data (Jiménez et al. 2009). Also, Chen et al. used the data collected by a probe vehicle equipped with a Carrier Phase Differential GPS aided inertial navigation system (INS) to reconstruct lane-level RDM (Chen et al. 2010). Knoop et al. argued the accuracy of the built-in GPS of the vehicles and smartphones is not enough for automated lane identification. So, they propose using Precise Point Positioning (PPP), which is a transitional technique between stand-alone GPS and Differential-GPS (D-GPS). Using this technique they could identify road lanes with 20 cm accuracy (Knoop et al. 2012, Knoop et al. 2017). Another example of using high-

precision GPS data is presented in a study by Zheng et al. They used an autonomous vehicle equipped with a GPS and high-precision (INS) (Zheng et al. 2019). Another study used an equipped car with Yaw rate sensor, wheel speed sensors and high accuracy GPS receivers to reconstruct maps for automated vehicles. Although the authors claim to have produced very accurate RDMs, their maps are not lane based (Jo and Sunwoo 2013). A similar study enhanced data sources with road imaging (Guo et al. 2016) which guarantees accurate lane identification. Contrary to the authors' claim, such a method is not only expensive, but its widespread use is also difficult. An advanced laser based distance measurement method known as LIDAR is also used for lane based RDM reconstruction solely (Wu et al. 2020) and by fusing data with GNSS and cellular data sources (Maaref et al. 2018). Joshi and James used a probe vehicle to collect data to reconstruct lane-level RDMs with high precision. They equipped their probe vehicle with a set of sensors including LIDAR, high-precision GPS, and an HD color camera. Then they combined the data from different sensors using a particle filter (Joshi and James 2015). In their study, Coifman et al. used a probe vehicle equipped with several sensors, including LIDAR, to collect data. Their main purpose was to extract empirical microscopic vehicular interactions. However, the dead-reckoning method used also allows to identify lanes with high accuracy and without any prior reference knowledge (Coifman et al. 2016a, Coifman et al. 2016b). The accuracy of such methods is very high, which can be essential for guiding autonomous vehicles. However, it is not applicable when using cheap and omnipresent low-precision trajectories like in our study. Indeed, the most important disadvantage of all of these efforts is that they are considerably expensive compared to our proposed method and due to the use of vehicles equipped with special sensors, it is not possible to reconstruct the RDMs on a large scale at low cost with the help of these methods.

3. Data collection and test network

Our test network consists of the E313 and R1 motorways between junctions Wommelgem and Antwerpen-Zuid in both driving directions near Antwerp, Belgium. The data was collected through passive observation of existing users of the Touring Mobilis and Flitsmeister smartphone apps of commercial traffic service provider Be-Mobile available for both iOS and Android operating systems. A geofence was defined around the motorway network (shown in Fig. 1 where the geofence is highlighted through a transparent blue mask). Upon entry of the geofence, trajectory data is recorded at 1 Hz frequency of each vehicle in which a user has this smartphone app installed. Because the drivers (or passengers) are completely anonymous and uninformed (other than their general consent, upon installation of the app, to collect and exchange anonymized data), there are no driving instructions, neither could vehicles be equipped with any supplementary antennas or other sensors. Thus, drivers are uninfluenced by the passive observation and decide independently on their driving lanes and maneuvers such as lane change and overtaking. Different users may place their smartphones anywhere in their vehicle by their personal choice. Trajectory data does not include the type and length of the vehicles. However, because of the diversity of Be-Mobile's customers, the database does not only include passenger cars rather a (possibly disproportionate) sample of different vehicle types. Trajectory data is sent online from the smartphone app to the Be-Mobile server and is available to the research team for analysis with a delay of a few hours.

Trajectory information including a dedicated unique ID for each vehicle (smartphone) and timestamp based on local time (Central European Time (CET)) and vehicle spatial location including latitude and longitude in the WSG84 coordinate system (EPSG: 4326), as well as speed and heading. Note that the location accuracy is unknown and varies broadly between trajectories depending on the smartphone GPS receiver quality, internal filtering, place inside the vehicle, interference with other signals, atmospheric conditions,



Fig. 1. The test network of this paper (the area covered by the transparent blue mask shows the geofence range). (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

multipath reflections etcetera. The information is stored at 1 Hz frequency. On weekdays, trajectories of on average 4200 vehicles are registered. Data storage started in December 2018 and continued beyond the time of publication of this paper. This paper uses the information of September 10 and 12, 2019, which were days without rain and traffic accidents in the test network (geofence).

4. Methodology

We propose a three-step algorithm that converts a set of trajectory data into a lane-level RDM. Each step uses universal methods that can be replicated with similar data on similar motorway networks. Fig. 2 gives an overview of the methods, which are detailed hereafter. The first step of the algorithm is network segmentation. At the end of this step, the position of the nodes in the network will be identified. We use these nodes to cut the motorway network into road segments. Each road segment is uni-directional and homogeneous in its entire length in terms of width and number of lanes. In the second step, the centerline will be constructed for each uni-directional road segment. Finally in the third step, the lateral distance of the trajectory sample points from the centerline of the road segments is the criterion for their clustering using a Gaussian model and identification of the lanes.

4.1. Step 1: network segmentation

At the first step, we identify nodes that divide the test network into homogeneous unidirectional road segments (in terms of the number of the lanes). As a result, not only nodes must be identified, but also sections where the number of lanes changes (increasing or dropping) without connecting to another link. Whereas the former is determined in this step of the algorithm, the exact determination of the latter is more complex, and it will be done in the third step of the algorithm, after calculating the width of each road segment. To identify nodes (link connection points) it is necessary to define the concept of bundle. Nodes are defined where two bundles of trajectories are joined together (merge or on-ramp) or separate (diverge or off-ramp). A bundle is defined as a set of trajectories in a road segment that follow the same path through the network; bundles are identified using the QuickBundles (QB) clustering algorithm (Garyfallidis et al., 2012). This clustering method firstly was proposed to use on 3D MRI streams. Its application to trajectory data is simply possible by considering the same value for the third dimension for all trajectory paths (for example considering zero as the third dimension of all trajectories). Fig. 3 shows bundles in an area of test network as well as two situations where bundles merge and diverge. The algorithm considers incidences, i.e., intersections of trajectory paths belonging to different bundles (due to lane changes and noise on the lateral position recordings). Sufficiently upstream of a diverge (resp. downstream of a merge), the number of incidences per distance unit is relatively stable; we call this stable quantity Q for a diverge (respectively Q' for a merge). The closer to a

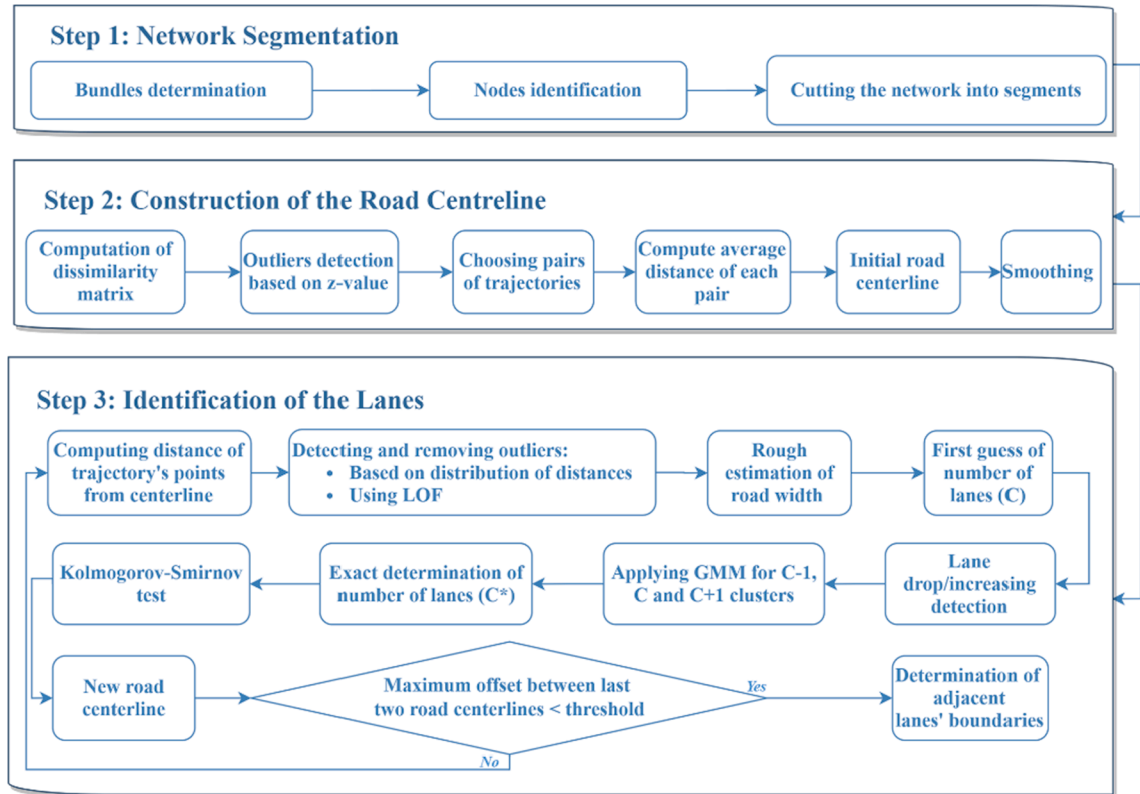


Fig. 2. Flowchart of the proposed algorithm.

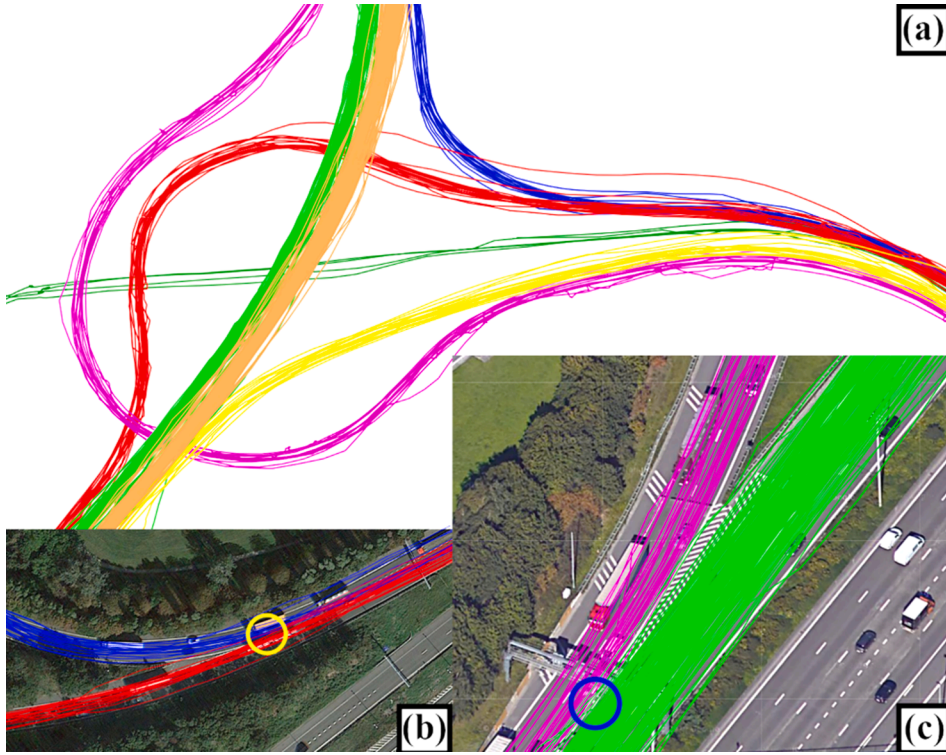


Fig. 3. (a) Trajectory bundles; (b) a node where two bundles of trajectories diverge; (c) a node where two bundles of trajectories merge.

diverge, the more the two bundles are separated, and the number of incidents tends from Q to zero. Likewise, upstream of a merge, the number of incidences tends from zero to Q' . In almost all cases in our data, the point where the last (resp. first) incidence occurs is a GPS data error. Two fractions of Q and Q' , let us call them f_Q and $f_{Q'}$ respectively, are the closest approximation to the points of the diverge (resp. merge) in the geometrical plan. Determination of f_Q and $f_{Q'}$ depends on several criteria and must be calibrated to use the proposed algorithm in this paper. The accuracy of the input data, the researcher's goal from the reconstruction of the RDMs, the accuracy that the researcher expects from the results, etc. affect these values. For the purpose of our study and based on the input data of this paper, the best values of f_Q and $f_{Q'}$ based on our data were equal to 0.035 and 0.027, respectively. These values could be considered as a good start for calibration in the future usage of the proposed algorithm to other sites or with other data.

4.2. Step 2: construction of the road centerline

The second step of our algorithm is the construction of the road centerline of the homogeneous, unidirectional segments. Previous

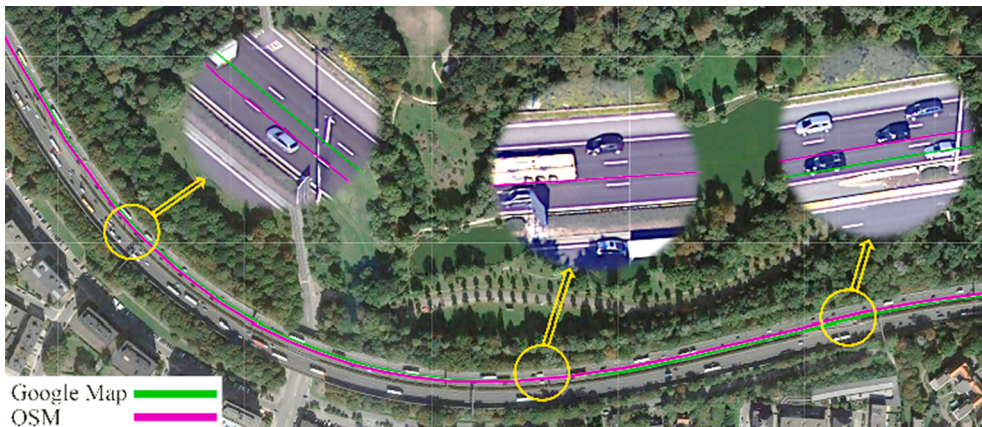


Fig. 4. Comparison of the road centerline in a motorway segment near Antwerp extracted from Google Map and OSM.

studies have used two main approaches. Some studies take the road centerline from existing RDMs or available maps, others determine it mostly based on density-based algorithms. The distribution of density of trajectories over the width of the road is however not constant, as traffic flow distribution over lanes is affected by lateral maneuvers at complex sections (such as merges, diverges, on/off-ramps, weaving sections). This may explain why, even if we may assume that large quantities of tracks were used, the road centerlines of Google Maps and OSM (shown in Fig. 4) are not consistent: they do not overlap, and their relative distance varies along the segment.

As an alternative we therefore developed a method based on the distance of the trajectory paths as curves. To measure the distance between trajectories, we used the Fréchet distance (Eiter and Mannila 1994). Three factors have been considered in selecting the trajectories' similarity measure. 1) Because different vehicles do not travel the length of a road segment at equal speeds (especially due to driver heterogeneity and differences in level of congestion at different times of the day), the number of trajectory points of different vehicles on the same road segment are not equal. Therefore, we should use a similarity measure that is capable of computing distance between trajectories with different number of sample points. 2) For the computation of the similarity matrix, temporal information of the trajectory sample points is not essential. What matters is the shape of the trajectory paths, regardless of the time difference between the sample points. Therefore, a measure that only considers spatial information is sufficient for the purpose of this study. 3) It is important to choose a parameter-free measure, as a result the algorithm would be more robust and calibrating the parameters will not become an issue (Sousa et al. 2020). Let us consider T_r as the set of all trajectory paths in each road segment r . The distance between all of the members of T_r is computed in a pairwise manner based on the Fréchet distance D^{ij} between each pair of trajectory paths (i and j); then a normalized dissimilarity score $0 < S_{FD}^{ij} \leq 1$ is computed based on these distances:

$$S_{FD}^{ij} = \frac{D^{ij} - \min_{ij}(D^{ij})}{\max_{ij}(D^{ij}) - \min_{ij}(D^{ij})} \quad \forall i, j \in T_r, i \neq j \quad (1)$$

Pairs with higher S_{FD}^{ij} are those that are laterally farther apart from each other compared to other pairs in a road segment. In order to avoid the use of error-full trajectories (e.g., too far off the road centerline due to the GPS error), we remove outlier trajectories in our dissimilarity matrix based on the z-score. The lateral error of this group of trajectories is sometimes up to 15 m, suggesting that they would be located in the refuge, shoulder, opposite road direction, etc. Using such obviously erroneous data will only cause bias in the construction of the centerline. To avoid this, we use a z-value equal to 1.96 to exclude trajectories with high error (e.g., projected off the road pavement). This threshold removes 4.56% of the most apart trajectory pairs from the continuation of this step of the algorithm.

Further, we will determine the centerline based on pairs of trajectories with high dissimilarity. The rationale behind this is that these pairs are most likely to be on the outermost lanes (rightmost and leftmost lane) across the entire segment. Hence, they mark the lateral extremities of the road (but thanks to eliminating excessive outliers first, not *beyond* the road) and the centerline must be approximately in the middle of the pair. By considering multiple dissimilar pairs, we get many samples of the centerline position, which allows us to reduce the approximation error of a single pair. Determining the best number of pairs of trajectories to reconstruct

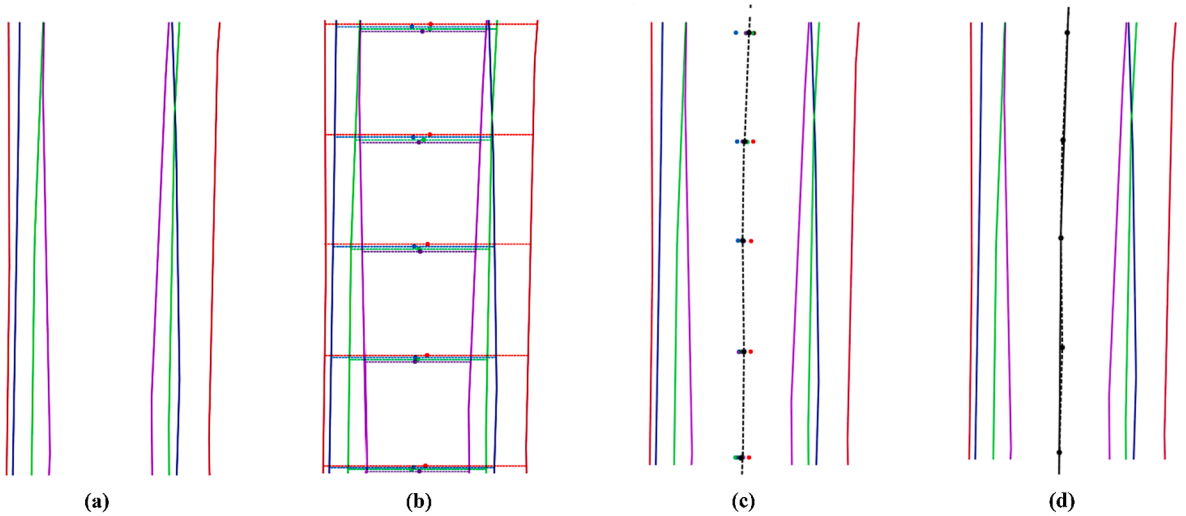


Fig. 5. (a) selection of highly dissimilar pairs of trajectories; (b) computing middle point samples; (c) determining the road centerline; (d) determining the smoothed road centerline.

the road centerline is challenging. Increasing the number of these pairs is statistically better, but too high numbers will lead the proposed algorithm to the same problem that density-based methods face (instability of density distribution of trajectory sample points across the road width, i.e., we would include pairs that have not driven on the outermost lanes over the full length, which biases the estimate). Selecting a small number of pairs, however, reduces the statistical reliability of the centerline position. We propose to not specify a number or fraction of all trajectories, but rather use a minimum dissimilarity threshold S^* . S^* is another calibration parameter of the proposed algorithm and its determination depends on three criteria. On one hand, S^* must be large enough to make sure that in multi-lane routes, pairs of trajectories that are in the same lane or adjacent lanes are not selected. On the other hand, S^* should not be chosen so large that a small number of pairs are selected for the next steps. The researcher must also make sure that the centerline is robust and is independent of the selected value of S^* . While pairs with an S_{FD}^{ij} greater than 0.944 are usually considered and eliminated as outliers based on the z-score, for our data set, selecting a threshold value of $S^* = 0.60$ ensures enough pairs of trajectories (farthest pairs) are selected on all road segments for the reconstruction of the road centerline. For other trajectory datasets and networks, this threshold may require specific recalibration. We have also considered the condition that each trajectory path can only be a member of one selected pair. A calibration method for S^* can be this way. Choose a large value for S^* (for example, 0.85) then gradually decrease S^* until the centerline become stable (decreasing from S^* will cause more trajectory pairs to be selected to construct the centerline). At the same time, the first of the above mentioned three criteria should not be violated.

The continuation of this step is described using Fig. 5. For better illustration in this figure, only four trajectory pairs are shown. In Fig. 5 (a) four pairs of trajectories each in their own color are chosen (for illustration purpose scales are not respected in this figure). Between the nodes identified in the previous step of the algorithm that bound the segment under consideration, all trajectory paths have approximately similar length. For each pair of trajectories at longitudinal intervals of Dx , their lateral distance from each other is computed and a midpoint is calculated. In this paper, we have determined the value of $Dx = 10$ meters. Determining Dx depends on the volume and quality of the available trajectory data. Very small values of Dx can prolong the algorithm run time and make the centerline somewhat noisy. Conversely, too large values of Dx will weaken the centerline robustness. These midpoint samples are shown over the dashed lines that represent the lateral distances of the trajectories of each pair in Fig. 5 (b) and (c) in the same color of the pairs. Then the average of the middle point samples is calculated (at longitudinal distances of Dx from each other; represented in Fig. 5 (c) with black points). The line connecting these points is the initial estimate of the centerline of the road segment (shown in dashed black line in Fig. 5 (c)). Finally, a smoothing step has been applied to avoid sharp curves at the sampling points. Let us define the set of all points at Dx interval as W . The smoother algorithm with an illustrative input and output are presented in Fig. 6 and the result of this step of the algorithm shown in Fig. 5 (d). Usually, the *while* loop in smoother code does not need to be evaluated more than 2–5 iterations for each road segment.

4.3. Step 3: identification of the lanes

The third and last step of the algorithm is the lane identification. In this step the first action is detecting and removing outliers in trajectories. To this end, the projected distance from the centerline of the road segment is calculated for each trajectory point. Then outlier points based on these distances are identified.

In this step of the proposed algorithm, a two-stage process for identifying and deleting outlier data is developed. The first stage is to logically remove obvious outlier trajectory points which are located on the opposite direction (side) of the road segment (which logically may not be the correct data). And the second stage is based on an adaptation of the local outlier factor (LOF) (Breunig et al. 2000). In this paper, we were looking for an outliers detection method that has fewer parameters and is more suitable for spatial data.

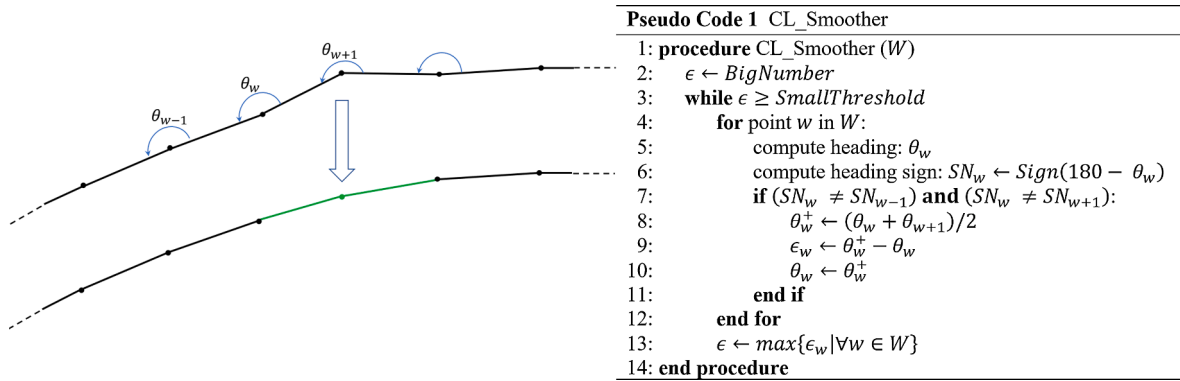


Fig. 6. Centerline Smoother pseudo code and schematic input and output.

In addition, the method should be capable of identifying both local and global outliers. LOF has only two parameters: the neighborhood size and the outliers' threshold. According to the literature LOF, while having the above-mentioned three characteristics, has shown good performance in evaluation on different datasets and works in a reasonable run time for big datasets compared to other methods (Schubert et al. 2014, Campos et al. 2016).

For the first stage, the distribution of trajectory points around the centerline of the road in each segment on different directions of two-way roads (in our test network all two-way roads are separated) has been used to obtain a preliminary estimate of obvious outliers ("obvious" because of physical infeasibility of the observation). The diagram of the distribution of trajectory points in two opposite directions of the two-way roads intersect at a point. Logically, the trajectory points beyond this intersection are outliers. Calculating the distance of this intersection point from the centerline of each road segment and discarding the data that is farther away on each side of the centerline of the road, ensures that at least the worst outliers are removed from the continuation of this step of the algorithm. In addition, it is used further for the adjustment of the LOF to increase its accuracy. A prototype of applying this part of the algorithm is shown in Fig. 7. For a better representation, in this figure, a cross-section of the test network is deliberately selected so that four peaks corresponding to four lanes on each road direction can be easily identified. Such clarity does not exist in most of the cross-sections of the network. This stage has been developed by the assumption of the symmetry of each road segment relative to its centerline. Therefore, even if the number of lanes in two opposite directions are not equal, this method can still be used. For road segments that have an emergency lane, this method is still valid, because first, the emergency lane is not important for our study purpose, and we only seek to estimate driving lanes. Second, trajectory paths that actually used the emergency lane (which is illegal in the absence of an emergency) should be rare exceptions in the dataset. And last but not least, this method is used as a calibration stage to better determine the LOF threshold (will be explained below). Therefore, even for ramps (which do not have the opposite direction), it is appropriate to use the obtained outlier boundary distance for road segments with the approximately same width.

Unlike many common outlier detection methods that assign an outlier or inlier binary tag to each sample point, LOF assigns a degree of being an outlier: the "local outlier factor". This is a measure of the isolation of each sample considering its neighbors, which is calculated as the density of neighbors around each sample relative to its nearest neighbors. Higher degree is assigned to points whose density has a significant drop with respect to its neighbors. To make the method more robust, a second criterion called "reachability distance" has been defined (Breunig et al. 2000). Let $k_d(p)$ be the distance of the sample point p to its k^{th} nearest neighbor and the set of these neighbor points is $N_k(p)$. And let $d(p, o)$ be the distance between sample points o and p . Consequently, the reachability distance $rd_k(p, o)$ between these two points and the local reachability density $lrd_k(p)$ of a sample point p in its neighborhood of size k are defined as:

$$rd_k(p, o) = \max\{k_d(p), d(p, o)\} \quad (2)$$

$$lrd_k(p) = 1 / \left(\frac{\sum_{o \in N_k(p)} rd_k(p, o)}{k} \right) \quad (3)$$

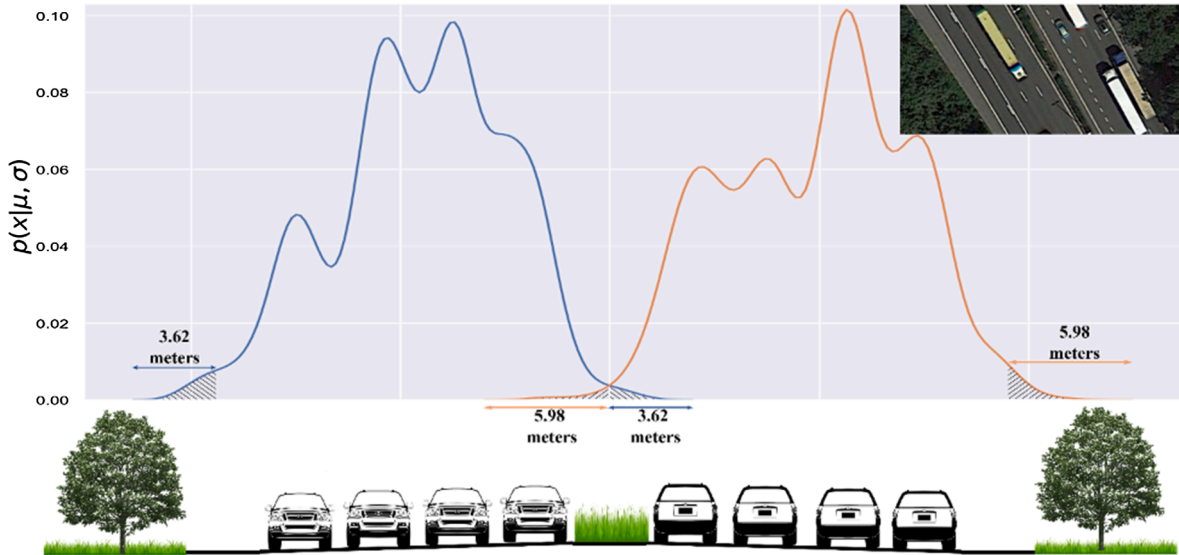


Fig. 7. Using distribution of trajectories around identified road centerline to detect outliers.

where $lrd_k(p)$ is the inverse of the average reachability distance of sample point p from its neighbors of set $N_k(p)$. Then the local outlier factor of sample point p is defined as Eq. (4):

$$LOF_k(p) = \frac{\sum_{o \in N_k(p)} lrd_k(o)}{k \cdot lrd_k(p)} \quad (4)$$

While LOF values less than 1 are definitely inliers (Breunig et al. 2000), it is up to the researcher to determine a suitable neighborhood size k and a precise threshold for LOF value (let us denote this threshold with LOF^*) beyond which samples are considered as outliers. The developers of the LOF algorithm have provided recommendations on how to determine k . In particular, it is emphasized that k must be greater than 10 (Breunig et al. 2000). In this paper, to determine the best value of k , we increase the value of this parameter gradually, until a further increase in k did not change the corresponding LOF value of each trajectory sample point. There is no need to repeat finding the best k for all network segments. For trajectory data collected from a single source with the same quality, the value of k is usually the same for all road segments. The LOF threshold should certainly be greater than 1. To find the most suitable threshold we take advantage of plotting the trend of changes in the LOF values concerning the trajectories' sample points' distance from the centerline (Fig. 8). Fig. 8 illustrates the LOF values for the rightmost lane of a four-lane road segment in the test network. Plotting similar diagrams for different road segments of the network shows a sudden jump in the LOF values that forms an elbow. The average LOF of points at the location of this elbow is the threshold we are looking for. This threshold is shown in Fig. 8 with a yellow arrow and it is corresponding to average LOF = 1.09, and 7.89 m distance from the road centerline. The red arrow represents the border of the outliers known by the method of Fig. 7. Green and blue arrows are added to guide the reader. The blue arrow indicates $LOF = 1$, which is closer to the road centerline than the actual edge of the road. And the green arrow shows the real edge of the road (which we checked in this case for validation but is in general not available). It is worth noting that without taking advantage of the method of Fig. 7 the elbow (the location of the yellow arrow) still forms but for example for the road segment of Fig. 8, it would lie at 8.42 m distance from the road centerline, i.e. failure to first remove the obvious outliers makes us more tolerant for some other outliers.

After removing outliers, based on the maximum distances of the remaining trajectory points on the left- and right-hand sides of the centerline, the maximum possible width of the road segment is obtained. As our test site is a motorway network, an initial guess about the number of lanes is given by dividing the maximum width obtained from the previous step by 3.75 m (let us call this guess C).

If the road segment is not homogeneous and a lane is added or dropped along it, the initial estimate of the road width along that road segment is not constant. In this part of the algorithm, the width of the road in each segment is controlled in consecutive cross sections at intervals of 10 m. Adding or dropping a lane on a motorway network means adding or dropping ± 3.75 m to the road width. This usually happens close to a node (motorway merge or diverge, or on-ramps and off-ramps) and rarely in the middle of a section. Therefore, a change of road width in a few consecutive cross sections far from a node is most likely caused by GPS error. We thus use

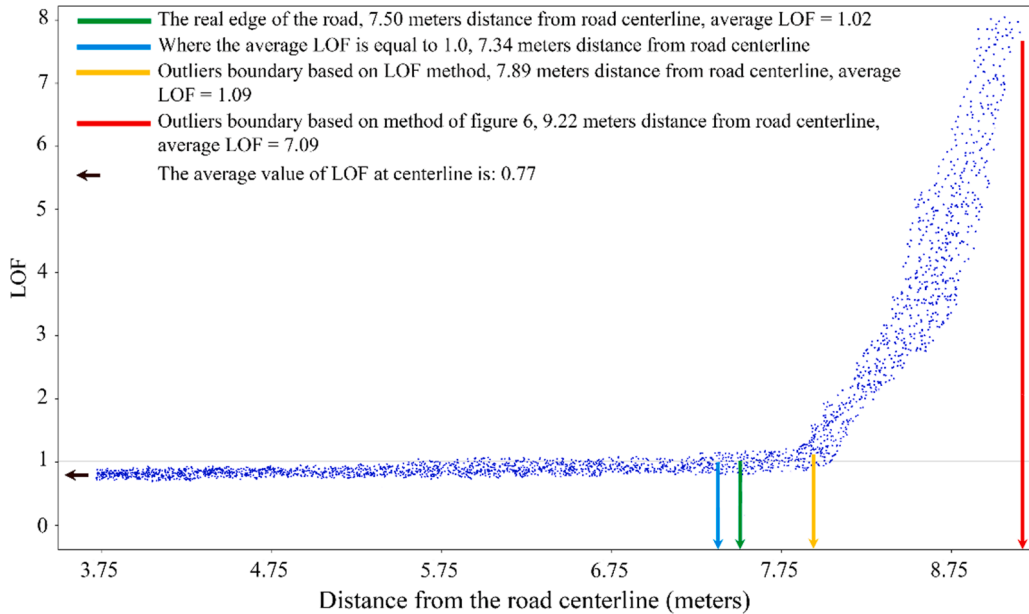


Fig. 8. Determining the LOF threshold to identify outliers in the most-right lane of a four-lane road segment in the test network. Trajectory sample points with LOF more than this threshold are considered as outliers.

two criteria to detect the addition or drop of a lane. The first criterion is that the change in the width should persist continuously over a length of at least 100 m; the second criterion is that it should be adjacent to a node. Whenever the two criteria hold, we need to determine accurately the starting point of the change of width. To this end, over a range of 15 m before and after the cross section where the width changes, the width of the road is recalculated at smaller intervals. We chose intervals of 0.5 m, but this can be adapted depending on the required accuracy.

With the road width known, we use a Gaussian Mixture Model (GMM) (Fraley and Raftery 1998) to identify lanes. The GMM method assumes that the data distribution is obtained from the combination of several normal distributions. At this stage it is impossible to test the normality assumption and we assume it to be true. But after estimating the clusters (lanes) we will check to what extent this initial assumption was correct. A normal mixed distribution $P(x)$ typically consists of a combination of $\mathcal{M}$ normal distributions:

$$P(x) = \sum_{m=1}^{\mathcal{M}} \pi_m \phi(x|\mu_m, \Sigma_m) \quad (5)$$

where ϕ is a normal distribution with mean μ and variance Σ and π_m is the weight of each distribution in the mixture. In addition, the probability distribution that has the highest probability of generating data points $\mathcal{X} = \{x_n : 1, \dots, n\}$ is $P(x) = \prod_{n=1}^N P(x_n)$, which is actually the likelihood function. We should find μ_m and Σ_m so that the likelihood function is maximized. A two-step algorithm that is known as *EM* algorithm is usually used in the literature to solve this maximization problem. But first we should compute the log-likelihood (*LL*) function because taking derivatives of *LL* function is more convenient:

$$\mathcal{L}(P(x)) = \sum_{n=1}^N \log\{P(x_n)\} = \sum_{n=1}^N \log\left\{\sum_{m=1}^{\mathcal{M}} \pi_m \phi(x_n|\mu_m, \Sigma_m)\right\} \quad (6)$$

In the step *E* of the *EM* algorithm a first guess or an already known value for μ_m and Σ_m is used to calculate the expected value of the *LL* function. For every data point x_n the probability that it is generated by the m^{th} component of mixture distribution calculated by Eq. (7):

$$\gamma(n, m) = \frac{\pi_m \phi(x_n|\mu_m, \Sigma_m)}{\sum_{m=1}^{\mathcal{M}} \pi_m \phi(x_n|\mu_m, \Sigma_m)} \quad (7)$$

Then, in the step *M* of the *EM* algorithm we maximize the *LL* function by taking derivative of the *LL* function with respect to μ_m and Σ_m and so obtain these parameters:

$$\mu_m = \frac{1}{\phi_m} \sum_{n=1}^N \gamma(n, m) x_n \quad (8)$$

$$\Sigma_m = \frac{1}{\phi_m} \sum_{n=1}^N \gamma(n, m) (x_n - \mu_m)(x_n - \mu_m)^T \quad (9)$$

$$\pi_m = \frac{1}{n} \sum_{n=1}^N \gamma(n, m) \quad (10)$$

The *E*-step and *M*-step should repeat until convergence.

Each of the normal distributions in a mixture has its variance, but their mixture also assumes a covariance between each pair of distributions. As we will discuss later on (in Section 5.1, Fig. 10), it is essential to assume the normal distributions are independent of each other, hence the covariance is assumed to be zero. Without this assumption, some unrealistic results may be obtained.

Also, given the range of the error in the GPS data, classification of the trajectory data with GMM can lead to the observation of kurtosis in clusters specially in those that are located in the sides of the road (See Fig. 10, Section 5.1). We have also noted that since the shares of the vehicles observed in the central lanes are typically larger than other lanes (at least in our test site), due to the density of observations, central clusters (lanes) may be computed too narrow. Consequently, the widths of the side lanes will be larger than 3.75 m and the widths of the central lanes will be less than the standard. So, we added a constraint to the GMM algorithm regarding the maximum and minimum acceptable standard deviation for clusters i.e., $0.90 \leq \sqrt{\Sigma_m} \leq 1.02, \forall m \in \mathcal{M}$ which are obtained based on the assumption of spreading of trajectory data points with $\sigma_m = 0.96 \pm \varepsilon_m$ meters around μ_m which is the center of each lane. Our investigations found that a value of $\varepsilon_m = 6$ centimeters is the smallest acceptable error in estimating the standard deviation of clusters and will lead to good final results. On the other hand, selecting smaller values can non-optimally increase the run time of the algorithm,

while the results will not improve significantly.

In addition, we consider $\pi_m = \pi_{m'}, \forall m, m' \in \mathcal{M}$. This means that we have assumed the share of all lanes (weight of all clusters) in the total traffic flow to be equal. This assumption is only helping for simplification and generality in case of lack of access to exogenous data. In addition, this is an assumption for starting the algorithm and is released after the first iteration. If $\pi_m, \forall m \in \mathcal{M}$ are borrowed from an exogenous data source, such as loop detectors, the proposed algorithm will work much better in terms of traffic metrics. In such a case, all $\pi_m, \forall m \in \mathcal{M}$ in the next iterations of the algorithm should be assumed to be fixed. This claim will show in [Section 5.3](#), [Table 4](#). As shown in [Table 4](#), even without obtaining π_m from exogenous sources, the proposed algorithm has independent performance with acceptable results.

An important contribution of this paper is that we classify trajectory data for a whole length of each segment of the road at once, and do not use vertical cross sections (perpendicular lines) whereas it is a conventional method in the literature. This ensures that the width and position of the lanes remain the same throughout each segment.

As the first guess of the number of the lanes (C) may be rough, we applied GMM for $C-1$, C and $C+1$ clusters in each road segment. We consider three criteria for the final decision to determine the true value of C , namely C^* . First, we used the Calinski-Harabasz Index (CHI) ([Desgraupes 2013](#)). This test will not only determine the C^* but also help to ensure the correct detection of lane drop and lane increasing sections. The limitation of CHI is that the number of clusters to be compared must be greater than 1, i.e., $C > 2$. Second, to compensate for this limitation, we also take advantage of the researcher's judgment about the width of the lanes. Next, we verify if scattering of the trajectory points on each lane indeed has a normal distribution or not by a Kolmogorov-Smirnov goodness of fit (KSG) test.

Then, the centerline found in step 2 is corrected by an offset derived from the identified lanes. For road segments with an odd number of lanes, the centerline is shifted towards the center of the middle lane; for segments with an even number of lanes, the centerline is shifted towards the mathematical intersect point of the identified normal distributions for two adjacent middle lanes. If the shifted centerline is offset by no more than a threshold (default: 1 cm), the algorithm is finished; otherwise, the whole algorithm of identifying of the lanes is repeated based on shifted centerline in a loop until the required shift is below the threshold.

Finally, we consider the boundary of two adjacent lanes the intersection point of the normal distribution diagram of their trajectory points. As a result, even if a data point is theoretically located at the end of the tail of the normal distribution of a lane, it is ultimately considered on the adjacent lane.

5. Evaluation of the proposed method

5.1. Results and discussion

In this subsection, the results of the proposed algorithm are presented; the next subsections are devoted to consistency analysis, validation, and robustness analysis, respectively. These results are based on the data collected on September 10th and 12nd, 2019. The algorithm found 20 nodes, which corresponds to the actual number of nodes based on the network geometry. Through these nodes, our test network is divided into 34 road segments. We also identified 14 cross-sections by changing the number of the lanes before and after them, through the algorithm. [Table 1](#) summarizes the results for detecting the position of the nodes and the centerlines of the road segments compared to Google Map data and the width of the lanes. We stated at the beginning of this paper that we are looking to build lane-level RDMs with the appropriate precision to provide a platform for extracting vehicle lateral maneuver data. That is why we criticized existing maps like Google Map. However, in this section, a comparison has been made with Google Map just because we want to compare our results with the results of other research, without making any judgments.

The position error of nodes identified by our algorithm is 2.65 m on average; for the purpose of comparison with the results of other studies in literature; Wang et al. reported errors between 5.39 and 12.02 m ([Wang et al. 2015](#)) and between 4.46 and 8.26 m in Tang et al. ([Tang et al., 2019a](#)). The lateral distance of the centerline deviated on average 0.19 m from that of Google Maps with a 0.04 m standard deviation. Compared to the literature, Costa et al. reported an average and standard deviation of 1.91 and 1.52 m, respectively ([Costa and Baldo 2015](#)). Ahmad et al. reported a minimum and maximum deviation of respectively 0.21 and 0.59 m ([Ahmed et al. 2015](#)). The difference in lane width estimated by our method compared to the 3.75 m standard road design is 0.07 m on average

Table 1
Overall results of the algorithm.

	Minimum	Maximum	Average	Standard deviation
Nodes position compared to Google Maps	0.64 m	4.89 m	2.65 m	0.55
Lateral distance of centerlines compared to Google Maps	0.03 m	0.41 m	0.19 m	0.04
Lanes width	3.58 m	3.94 m	3.82 m	0.07

(0.07 m standard deviation). In comparison, Tang et al. (Tang et al. 2016) found errors of 0.35 m. On the other hand, Knoop et al. (Knoop et al. 2017) approximated the lane width with an average error of 0.05 m only; however, their estimate was based on precision point positioning (PPP) data, which is much more accurate than the smartphone GPS data in our study.

Fig. 9 represents examples of the road centerline for different locations of a four lanes segment in the test network. The significance of the obtained centerline is its consistency throughout its length, even though it is constructed at low cost, solely based on a small amount of smartphone GPS data. One limitation of the resulting centerline is however that it deviates slightly to the curve direction as a result of centrifugal bias of GPS data registered by smartphones (due to their relatively simple internal filtering) especially in high speed, curvy segments. Although this error can be partially corrected by the simultaneous use of high-speed and low-speed trajectories in curves, still, especially in the middle of the curves, sometimes more than one meter deviation to the outside of the curve is observed. For some applications, this error may be prohibitive, and remedies need to be found. In our case, we use the RDM to match GPS trajectories on and analyze lane changes (among others). Then, the same (centrifugal) bias is in the GPS track and in the RDM derived from the data, so that the bias cancels out while positioning relatively the track on the map.

As explained earlier, to obtain realistic results, we considered two assumptions: (i) zero covariance of the clusters in the Gaussian mixture, and (ii) a minimum and a maximum variance for each cluster. Fig. 10 shows a comparison of the effect of applying and not applying these assumptions on the results of the GMM algorithm in a four-lane segment of the test network. Panels “a”, “b” and “c” are all calculated for the same data in this figure. In panel “a”, neither of the two assumptions is considered. In section “b” the assumption of independence of clusters is considered, but no restriction on the acceptable minimum and maximum standard deviations is applied. In Section “c”, both assumptions are considered. Only applying both assumptions could guarantee clustering results truly representing road lanes.

It is not possible to provide detailed results for all 34 road segments of our test network in a paper. Therefore, we have selected three segments of the network as samples. Segment “a” is a three lanes ramp from the R1 outer ring of Antwerp to E313 eastbound. Segment “b” is approximately 600 m of R1 outer ring (i.e., northbound direction) with five lanes and finally segment “c” is approximately 1600 m of E313 westbound with four lanes. Table 2 presents lane identification results in these segments. Overall, in 77.6% of cases, KSG results confirmed the estimated clusters follow normal distribution at significance level $\alpha \leq 0.05$, in 13.6% of cases at level $\alpha \leq 0.15$,



Fig. 9. Calculated road centerline in a segment in the test site.

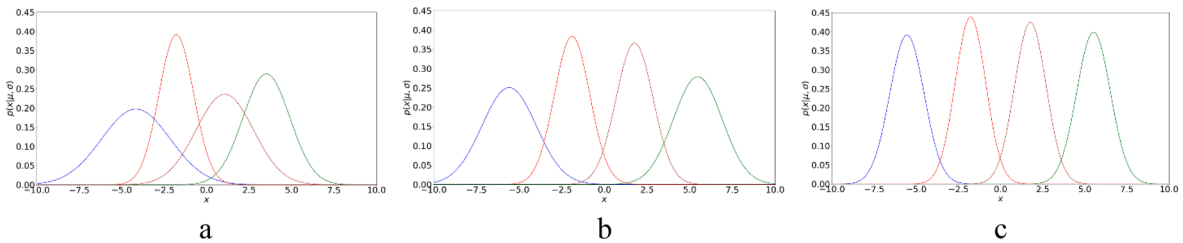
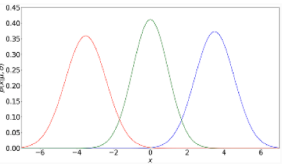
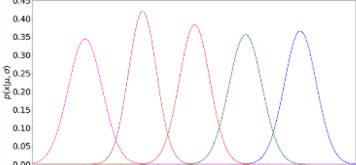
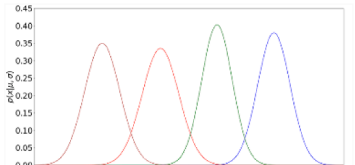


Fig. 10. Comparison of the effect of applying and not applying of the assumptions of this paper on the results of the GMM algorithm.

Table 2
Sample results of the lane identification step.

Location in the test network	  		
Probability distribution of the clusters	  		
Separation point of clusters	-1.81	1.91	
Lane width	3.80	3.72	3.87
Results of KSG test	0.042	0.041	0.063
	3.81	3.70	3.69
	3.77	3.82	
	3.85	3.68	3.69
	3.81		
	0.046	0.042	0.032
	0.040		

and in 8.8% of cases the KSG test was accepted at $\alpha \leq 0.2$. Fig. 11 shows the estimated clusters of trajectory points in three selected segments. And finally, Fig. 12 provides a view of the estimated lanes projected on an aerial Google photo for three different locations in the test network.

5.2. Analysis of the results: connectivity and compatibility

We claim the reconstructed lane-level RDM is a connected and compatible network. Our definition of network connectivity is that lane separators of successive road segments smoothly connect at the nodes without discontinuity. Therefore, small lateral distances between lane separators at nodes guarantee network connectivity. Of course, this is somewhat a compromise. The quality and volume of the input trajectory data and the purpose for which our proposed algorithm is used are important for judging the network connectivity. As a result, we cannot guarantee with other definitions and input data with any degree of quality, the output is essentially a connected network. We can only confirm that for the network we studied and our purpose, connectivity was sufficiently smooth. Certain applications of the proposed algorithm may require smoothening the small remaining discontinuities of lane separators over some transition region around the nodes. Also, for some applications, such as navigation of autonomous vehicles, there may be other definitions of connectivity that have not been addressed in this paper.

By compatibility, we mean that at the nodes' locations, the width of the corresponding lanes in successive road segments are compatible. This concept is illustrated by Fig. 13 (This figure has been prepared for the purpose of showing the concept only and the scales have not been conserved in it). Also, Table 3 shows the results of the evaluation of these two criteria.

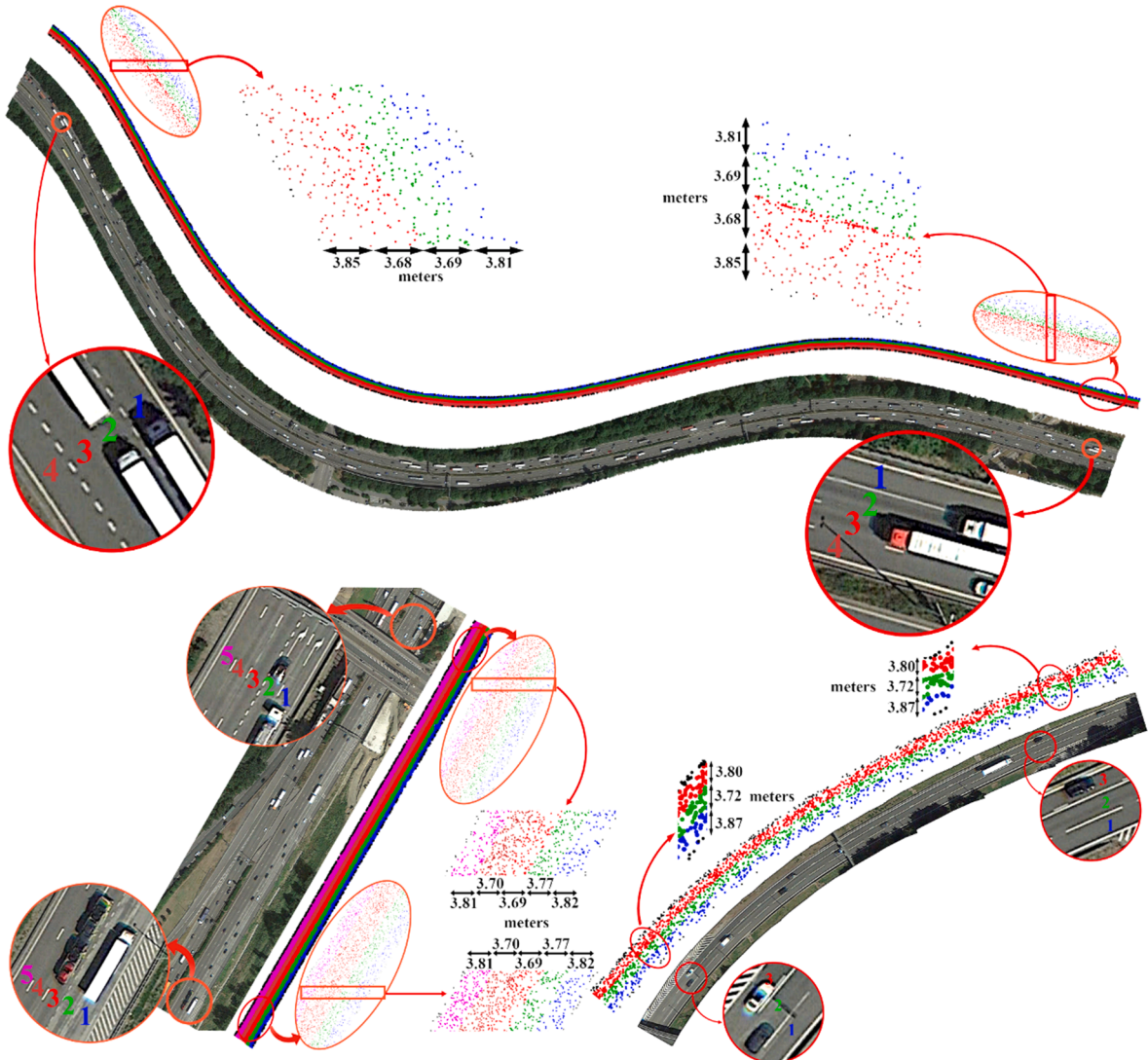


Fig. 11. Identified clusters of trajectory points based on the proposed modified GMM.



Fig. 12. Estimated lanes projected on an aerial Google photo for three different locations in the test network.

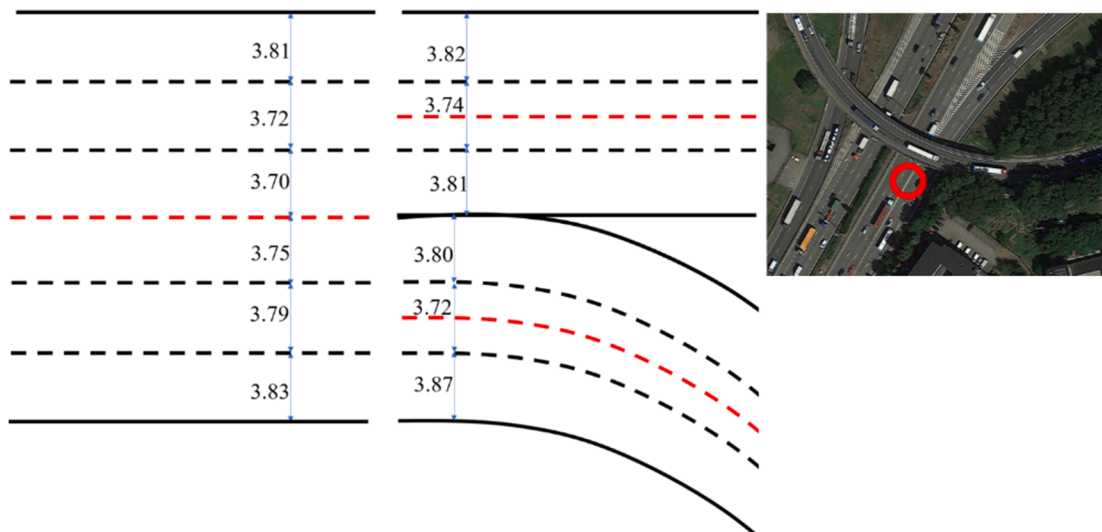


Fig. 13. An example of the connectivity of road segments when a 5-lanes segment reaches a diverge point.

Table 3

Evaluation of connectivity and compatibility of constructed lane-level RDM.

Criteria	Minimum	Maximum	Average	Standard deviation
Connectivity (lateral distance of lane separators in successive road segments in meters)	0.00	0.23	0.09	0.0203
Compatibility (difference in width of corresponding lanes in successive road segments in meters)	0.00	0.18	0.07	0.0194

The results of [Table 3](#) show that according to the definitions presented in this paper, the reconstructed lane-level RDM is connected and compatible. The average and even maximum difference in the width of consecutive lanes is very low for our study purposes. Also, the maximum lateral distance between lane separators of the corresponding lanes in successive road segments is only 23 cm (which is narrower than the lane marker painting). This slight difference makes the reconstructed lane-level RDM a suitable platform for extracting vehicles' lateral maneuver data.

Various studies have confirmed that at curves as well as at side lanes in a tangent, drivers have a slight lateral deviation from the center of the lane (depending on curvature, driving speed, roadside condition, etc. at various sources report this deviation approximately between 25 and 60 cm) ([Bella 2013](#), [Mecheri et al. 2017](#)). Therefore, our estimated lane may exhibit a bias compared to the actual physical lane marking. First of all, we limit this bias somewhat by imposing constraints on the variances in our GMM, but this does not render our estimate entirely immune to some lateral deviation. More importantly, however, we actually *prefer* to not entirely remove this bias as our goal is to create maps that are as consistent as possible with the trajectory data collected by smartphones. When later (in forthcoming work) we will analyze trajectories with respect to this reference map to infer information on lane choice and lane changes, we care about the functional lateral reference position more than about where the actual physical lane marking is. For example, if in a curve the average vehicle (that has no intention to change lanes) cuts the corner and is observed driving on the painted markings, then our functional lanes preferably follow that average vehicle rather than the physical marking. Rephrasing: we prefer the same bias in our reference and in our track, because this allows the best estimate of the track's *relative* position on the road, i.e. in which lane it is driving, or between which lanes it is changing. Note, however, that for other purposes the legal, physical lane marker may be needed as a reference, in which case we discommend using our method.

5.3. Validation

The most important impetus for us to perform this study was to obtain lane-based traffic data, which inspired us to validate through a traffic-based comparison. We chose as comparison criteria the average speed of each lane and average fraction of the total traffic flow per lane both in 15 min intervals based on trajectories and loop detectors data, at the location of the loop detectors (hereafter called measurements). We then compute root-mean-square error (RMSE):

$$RMSE = \sqrt{\frac{\sum_{i=1}^n (MT_i - MD_i)^2}{n}} \quad (11)$$

In this equation MT and MD are measurement based on trajectories and loop detectors respectively and n is the number of time intervals per location. We performed the validation for two four-hour intervals in the morning peak (from 7:00 to 11:00) and in the evening peak (from 14:00 to 18:00), yielding $n = 32$ time periods per loop detector. RMSE has the same unit as the variables so its absolute value is not proper for comparison. We normalized RMSE by a normalization factor NF . For the traffic volume share, we estimate NF as the expected order of magnitude, which in case of homogeneous distribution equals 1 divided by the number of the lanes; for speed NF is simply based on the average of speed per observations:

$$NRMSE = \frac{RMSE}{NF} \times 100 \quad (12)$$

It is worth noting that the comparison of the average traffic speed per lane based on the loop detectors' data and trajectory data (after lane-level RDM reconstruction) is only statistically significant if the speed difference of adjacent lanes is so large that the speed estimation errors (errors reported in [Table 4](#)) do not affect it. In addition, the researcher must ensure that an external factor such as loop detector failure or errors in the trajectory data collection application does not cause a systematic error in recording speed information. Our investigations (through a probe vehicle carrying nine smartphones simultaneously) show that there is no such systematic error in the input data of this paper, and the difference in speed based on loop detectors and trajectory data is only white noise.

In [Section 4.3](#) we discussed that two assumptions can be made for the weight of clusters ($\pi_m \forall m \in \mathcal{M}$) in lanes estimation. First, it assumed weights to be equal in the first iteration and then determine them in the EM algorithm. Or import weights from an exogenous source and assumed them to be fixed throughout the EM algorithm. The results of these two assumptions can be compared through columns 5 and 6 of [Table 4](#). While the use of exogenous data increases the accuracy of the proposed algorithm, the results are still of high quality without access to such a source.

The validation results for the same segments as shown in [Table 2](#) are summarized in [Table 4](#). One main reason for higher error 0 in the traffic volume lane share is the centrifugal deviation of the GPS records in the road curvatures, which increases especially with increase of the speed.

A more detailed examination of the validation results of the proposed method reveals that not only the error ranges in determining

Table 4

The validation results.

Road Segment	Number of per lane loop detectors	n*	Normalized RMSE lanes' average speed	Normalized RMSE fraction of traffic flow in lanes (determine π_m in EM algorithm)	Normalized RMSE fraction of traffic flow in lanes (exogenous π_m)
a	$2 \times 3 = 6$	183	3.31%	9.55%	1.87%
b	$2 \times 5 = 10$	307	4.87%	12.26%	2.02%
c	$5 \times 4 = 20$	622	4.12%	10.04%	1.61%

* On various days because of some reasons such as maintenance, loop detector data may not be recorded continuously during the study period.

Table 5

Comparison of traffic metrics computed based on trajectories and loop detectors' data.

Lane Number*	Average speed of lanes (Km/H)		Share of lanes from total traffic flow (percentage)	
	Based on trajectories	Based on loop detectors	Based on trajectories	Based on loop detectors
1	61.39	62.82	8.1	6.8
2	51.38	52.32	26.8	28.8
3	60.62	57.86	29.6	31.2
4	69.55	66.07	35.5	33.2

* Lane numbers are increasing from right to left in the flow direction.

the traffic metrics are low and, on a scale, suitable for traffic studies, but also the magnitude order of these two traffic metrics in the corresponding lanes based on trajectories and loop detectors' data are the same. To represent this feature, Table 5 shows the average speed of traffic flow and the share of lanes from the total traffic flow in a one hour period from 7:00 to 8:00 AM on a working day in September 2019 for a loop detector located in the coordinates (51.210797, 4.466583) in the road segment "c". As can be seen, the magnitude order for both traffic metrics are in the same sequence for corresponding lanes based on both sources of data.

5.4. Robustness analysis

One of the most important factors in choosing the proposed method for use in similar studies and other networks is its robustness relative to the minimum number of sample trajectories that is required for a reliable application of the method. For the sake of convenience, the results of the robustness analysis are also reported for the same segments as Table 2 and Fig. 11. The road's centerline and the lanes in three segments "a", "b" and "c" are identified with 889, 1783 and 1421 trajectory paths which have passed through these segments, respectively. In order to analyze the robustness of the proposed algorithm, the number of samples has been reduced with 10% steps and the results are presented in Table 6.

As the results of Table 6 show, depending on the geometric design and the number of lanes, approximately 500–550 samples of trajectories (60%, 30% and 40% of the samples for section "a", "b" and "c" respectively) are sufficient to apply the proposed algorithm reliably. The proposed method is more robust in term of correctly detecting the number of lanes compared to other criteria. In contrast, the sensitivity of increasing error in identified lane width relative to decreasing sample size is greater than the other two criteria.

Table 6

Robustness analysis of the proposed algorithm based on the trajectories sample size.

Road Segment	Criteria	100%*	90%*	80%*	70%*	60%*	50%*	40%*	30%*
a	Relative position of the road centerline**	–	1.6	5.5	10.4	27.6	72.5	114.5	245.8
	Number of lanes	3	3	3	3	3	3	2	2
	Width difference with standard lane***	8.7	9.9	14.2	21.3	29.5	38.5	50.2	64.4
b	Relative position of the road centerline**	–	<0.5	<0.5	<0.5	1.4	6.3	16.2	29.4
	Number of lanes	5	5	5	5	5	5	5	5
	Width difference with standard lane***	7.6	7.8	7.6	8.9	13.3	18.9	24.9	28.2
c	Relative position of the road centerline**	–	<0.5	<0.5	1.1	2.9	8.8	22.1	60.7
	Number of lanes	4	4	4	4	4	4	4	3
	Width difference with standard lane***	10.7	9.9	12.2	13.5	15.9	19.9	27.9	43.69

* Percentage of full sample.

** This criterion is calculated as the average change in the position of the road's centerline relative to the situation of using the 100% sample and reported in centimeters.

*** Computed as average of absolute difference between width of estimated lanes and standard 3.75 m lane in centimeters.

6. Remarks and conclusion

In this paper, we present a three-steps algorithm for construction of the lane-level routable digital maps (RDMs). Our main motivation for the paper was to provide a framework for the extraction and establishment of lane-based traffic databases from floating car data (trajectories). The results presented in this paper show that the proposed algorithm can determine the nodes, centerline, and road lanes in a motorway network. Most of the previous studies are only able to produce one or two components of the results of this algorithm. One of the most important contributions of this paper is the stability of the width of the estimated lanes along each road segment. The way that the GMM method is applied in this paper ensures that the error caused by fixed obstacles (such as bridges, etc.) in the trajectory data does not cause bias in the estimated lanes. Comparing the results of our algorithm with previous studies using comparable low-precision GPS data shows that the method of this paper gives reliable results. Another important achievements and contributions of this paper is that these results are obtained with a low-cost method of data collection, so that it can be easily replicated, and it is applicable in large-scale. Also, the results of robustness analysis of this method show that having 500–550 trajectories is enough to construct a reliable lane-level RDM.

In addition, with respect to traffic measurements such as speed and lane share from total traffic flow, we obtain close approximations solely based on trajectory data with an acceptable range of error (in average about 4% in term of speed and approximately 10% in terms of lane share from traffic flow). Another important contribution of this paper is its traffic validation. To the best of our knowledge, this is the first time that traffic metrics have been used to evaluate map-reconstruction methods. The results of traffic validation and the compatibility and connectivity features (considering the represented definition) of our method show its ability and reliability to be used in traffic studies. As explained in the introduction, the data gap of large-scale vehicle lateral maneuvers is strongly felt. The ultimate goal of our research is to enable the extraction of this data and then use it in the management of complex motorway networks. The lane-level RDM reconstructed base on the proposed method is considered as a suitable platform for extracting this data. Consequently, this algorithm provides a clear perspective of the possibility of achieving the ultimate goal of our research. This is a necessary step in enabling data fusion from multiple sources, among which trajectory data collected by smartphones (cheap and large-scale trajectory data), more accurately and to estimate traffic state more precisely.

Despite these important achievements, the proposed method can still be improved and upgraded. This method has been proposed and tested for motorway networks, a good direction for the future development of this method is to adapt it to the urban road network. In that context, however, it is indispensable to separate GPS tracks from cyclists, pedestrians, and any other off-road user of data collection smartphone app. Another topic for future research is the fact that in the current method, the resulting ‘functional’ centerline and lane markings may be systematically biased compared to the physical infrastructure because the method builds solely on trajectory data collected by smartphones. This bias can be caused by the GPS positions (due to internal filtering) being biased outward in sharp horizontal curves or because the lateral position of the driver population may systematically deviate from the physical center of the lane. If the application of the resulting map requires the physical road geometry rather than the functional one, such biases need to be corrected. This may necessitate the fusion of trajectory data with other physical references.

CRediT authorship contribution statement

Mohammad Ali Arman: Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Data curation, Writing - original draft, Writing - review & editing, Visualization. **Chris M.J. Tampère:** Conceptualization, Methodology, Validation, Investigation, Resources, Data curation, Writing - original draft, Writing - review & editing, Supervision, Project administration, Funding acquisition.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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